# Prototypical Contrastive Learning for Multi-Policy Trustworthy Reward Modeling

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## Abstract

Reward models (RMs) trained for reinforcement learning from human feedback face fundamental challenges when enforcing multiple trustworthy AI policies simultaneously. The preference data inherently follows a mixture distribution derived from distinct policies. The traditional approaches treat the data as a unified binary problem. This mismatch causes RM fail at distinguishing the underlying policy origins of preferences, leading to systematic over-refusal on benign queries while missing genuinely policy-violating content. We propose Prototypical Contrastive Learning based Reward Modeling (PCL-RM), which explicitly models the multi-policy structure of data by learning distinct prototypes. Experiments across multiple trustworthy AI benchmarks demonstrate substantial improvements over standard reward modeling in both reducing false refusals on benign queries and detecting policy-violating content, with more stable downstream RLHF training. Analysis reveals that a significant portion of learned prototypes exhibit high policy-specificity, demonstrating that our method discovers meaningful policy structure without explicit supervision.

## 1 Introduction

Reward Models (RMs) play a crucial role in aligning Large Language Models (LLMs) with human values and preferences through reinforcement learning. Recent advances in Reinforcement Learning from Human Feedback (RLHF) have shown promising results in implementing guardrails for LLMs using reward models trained on safe and unsafe data pairs (Dai et al., 2023). However, fundamental challenges emerge when attempting to enforce multiple complex policy constraints simultaneously within a unified safety framework.

A key insight driving our work is the inherent complexity of the trustworthy AI preference data distribution. While the data is typically presented as binary—'safe' versus 'unsafe' responses—the underlying distribution is actually a mixture derived from multiple distinct policies. Each model's safety specification reflects its builder's definition of safety rules, which varies across organizations and use cases. Traditional reward modeling (Ouyang et al., 2022; Bai et al., 2022) approaches using Bradley-Terry (BT) model (BRADLEY and TERRY, 1952) based preference learning treat this distribution in a unified manner, combining all data as preference pairs without distinguishing their underlying policy origins.

This unified modeling paradigm suffers from critical limitations. First, when trained on preference pairs from diverse policy sources, models struggle to disentangle competing signals, often learning to rely on superficial correlations rather than developing nuanced understanding of distinct policy requirements (Hong et al., 2025).Second, this approach leads to systematic over-refusal behavior: models trained to maximize a single reward signal become overly conservative, deflecting benign queries that superficially resemble policy-violating content while simultaneously failing to capture genuinely problematic responses that violate policies in subtle ways (Hong et al., 2025; Ye et al., 2026).

In this work, we address these fundamental limitations through Prototypical Contrastive Learning (PCL) for reward modeling, which explicitly models the multi-policy structure of trustworthy AI data. Rather than learning a single scalar reward function, our approach learns multiple prototypes representing distinct patterns of policy-compliant and policy-violating behaviors. These prototypes

serve as centroids in a structured embedding space organized as a hyperspherical manifold, where samples cluster around prototypes corresponding to their underlying policy origins.

Our contributions are twofold. First, we propose PCL for Reward Modeling (PCL-RM), which explicitly models this multi-policy structure through learned prototypes organized in a hyperspherical embedding space, combining preference learning with contrastive objectives to achieve both discriminative performance and interpretable structure. Second, we validate our approach across six trustworthy AI benchmarks, demonstrating substantial improvements in both reward model accuracy and downstream RLHF policy quality, with particular gains in reducing false refusals on benign queries while improving detection of policy-violating content. Analysis of learned prototypes reveals that our method discovers meaningful policy-specific structure without explicit supervision, with a significant portion of prototypes exhibiting high specificity to particular policy dimensions.

## 2 Related Work

Recent work has explored integrating contrastive learning into reward modeling to address key challenges in RLHF. Contrastive reward approaches have been shown to improve reward model robustness, reduce variance in policy optimization, and better calibrate according to task difficulty. Shen et al. (2024) demonstrate how contrastive rewards can improve RLHF by penalizing reward uncertainty and improving robustness in PPO. Similar to our approach, Chen et al. (2024) integrate contrastive learning into reward model training to enhance the model's understanding of differences among responses. However, these works focus on general reward modeling approaches, whereas in this work, we consider safety reward modeling that has a more discriminative nature and requires learning policy-specific representations to capture distinct compliance patterns across multiple RAI policies.

Prototypical Contrastive Learning (PCL) unifies clustering and contrastive learning by learning representations that capture both local smoothness and hierarchical semantic structure. The standard PCL framework has been successfully utilized in unsupervised learning for discovering underlying clustering structures in the embedding space (Li et al., 2021). Lu et al. (2024) further extended this approach to mixture of prototype representation spaces for out-of-domain detection. The prototypical framework naturally aligns with hyperspherical learning, as prototypes can serve as mean directions for von Mises-Fisher (vMF) distributions on the unit sphere, providing a principled probabilistic foundation for representation learning. Building on these foundations, our work combines vMF mixture models with policy-specific prototypical contrastive learning for multi-policy safety reward modeling, enabling interpretable and robust alignment across diverse RAI dimensions while maintaining the discriminative characteristics essential for safety applications.

## 3 Methodology

Our approach is motivated by observations about the nature of multi-policy sourced preference data. The training data typically consists of paired responses to both sensitive ("red") and benign queries, where appropriate responses often involve deflection strategies for potentially harmful content. Within each policy domain, these preference pairs exhibit highly discriminative characteristics.

While standard pairwise contrastive learning approaches enforce discriminative structures, PCL offers enhanced interpretability by learning representative prototypes for chosen and rejected responses. In this framework, policy-compliant samples are drawn towards their corresponding policy prototypes, while policy-violating samples are pushed into distinct, well-separated clusters. This geometric structure naturally enforces policy boundaries while maintaining interpretability through prototype-based reasoning.

### 3.1 Formulation

Denote $X$ and $Y$ as the prompt text and response text space. Given a prompt $x \in X$, we aim to score the harmlessness of a response $y \in Y$ with respect to multiple defined policies. Let $\mathbf{P} = p_1, ..., p_M$ be the set of $M$ distinct policies. Each policy $p_i$ defines its own distribution over compliant and policy-violating response. We assume the overall data distribution $\mathbf{D}$ of

responses follows a mixture model:

$$\mathbf{D} = \sum_{i=1}^{M} \pi_i \mathbf{D}_i, \quad (1)$$

where $\mathbf{D}_i$ is the distribution associated with policy $p_i$ and $\pi_i$ are mixture weights. In RM preference learning, each $\mathbf{D}_i$ is further decomposed into policy-compliant $\mathbf{D}_i^+$ and policy-violating $\mathbf{D}_i^-$ components. More specifically, we are given a set of $(x, y^+, y)$ tuple where $x$ is the input prompt and $y^+$,$y$ denotes the chosen and rejected responses respectively. In the trustworthy AI domain, the preference pairs exhibit high discriminative characteristics specific to each policy, i.e., $y^- \sim \mathbf{D}_i^-$ and $y^+ \sim \mathbf{D}_i^+$. Given this mixture distribution structure, our goal is to learn a reward function $R : X \times Y \to \mathcal{R}$ that effectively captures the compliance requirements of all policies.

### 3.2 Prototypical Contrastive Learning for Reward Modeling

Building upon the mixture distribution framework, we propose using PCL (Li et al., 2021) to learn a structured representation space. Instead of directly learning a scalar reward, we learn two sets of prototypes $\{\mu_1^c, ..., \mu_K^c\}_{c=\{+,-\}}$ for policy-compliant and policy-violating behaviors that represent the underlying policy structure. We model the distribution $\mathbf{D}^+$ and $\mathbf{D}^-$ (aggregated across all policies) using mixtures of von Mises-Fisher (vMF) distributions:

$$p(\mathbf{z}|\mathbf{D}^+) = \sum_{k=1}^{K} w_k^+ \exp\left(\mu_k^{+T}\mathbf{z}/\tau\right) Z_D, \quad (2)$$

$$p(\mathbf{z}|\mathbf{D}^-) = \sum_{k=1}^{K} w_k^- \exp\left(\mu_k^{-T}\mathbf{z}/\tau\right) Z_D, \quad (3)$$

where $\mathbf{z} = e_\theta(x, y)$ is the embeded representation of the input-response pair produced by encoder. $\{\mu_k^+, \mu{-}_k\}_{k=1,...,K}$ are the learned prototypes for prefered and un-prefered responses repectively. $w^+, w^-$ are mixture weights satisfying $\sum_k w_k^c = 1, c \in \{+, -\}$, $\tau$ is a temperature parameter. $Z_D$ is noramlization term. While the data distribution is governed by $M$ distinct policies, we learn $K$ prototypes per preference class to capture the underlying structure, where $K$ is a hyperparameter that may be chosen independently of $M$.

Our objective is to obtain a reward function: $R_\theta(x, y) = f_\theta(\mathbf{z}) = f_\theta(e_\theta(x, y))$, where $f_\theta$ is a learned reward head (typically a linear layer) that maps the presentation $\mathbf{z}$ to a scalar reward. The key difference from traditional approaches is that the representation $\mathbf{z}$ is learned through PCL, which structures the embedding space according to the underlying policy mixture. This structured representation enables the reward model to better capture policy-specific characteristics and maintain distinct decision boundaries for different types of policy violations.

### 3.3 PCL Training Objective

The training objective combines three components that work together to shape the representation space while learning preferences:

$$\mathcal{L} = \mathcal{L}_{BT} + \lambda_1 \mathcal{L}_{MLE} + \lambda_2 \mathcal{L}_{proto-contra}. \quad (4)$$

**BT based preference learning loss**: The reward model is trained to minimize the negative log-likelihood loss function:

$$\mathcal{L}_{BT}(\theta) = -E_{(x,y^+,y^-)\sim\mathbf{D}}[\log \sigma(R_\theta(x, y^+) - R_\theta(x, y^-))],$$

where $\sigma$ is the sigmoid function. This maintains the basic preference learning capability across paired responses.

**Prototype Assignment through MLE**: To encourage samples to cluster around appropriate prototypes based on their preference labels, we minimize:

$$\mathcal{L}_{MLE} = -\sum_c E_{(x,y,c)\sim\mathbf{D}^c}\left[\log p(\mathbf{z}|\mathbf{D}^c)\right]$$
$$= -\sum_c E_{(x,y^c)\sim_{D^c}}\left[\log \sum_{k=1}^{K} w_k^c \exp(\mu_k^{cT}\mathbf{z}/\tau)\right],$$

where $c \in \{+, -\}$.This loss encourages the encoder to map responses to regions near their corresponding preference-class prototypes.

**Prototypical Contrastive Loss**: To shape the prototype space and maintain clear separation between compliant and non-compliant clusters, we apply a contrastive loss on the prototypes themselves:

$$\mathcal{L}_{proto-contra} = -\sum_c \sum_{k=1}^{K} \log\left[\exp \mu_k^{cT}\mu_k\right], \quad (5)$$

This loss encourages prototypes within the same preference class to maintain consistent representations while maximizing separation between chosen and rejected prototypes.

By minimizing the loss function in Eq (4), the RM learns the discriminative preference and the underlying multiple prototypes centered embedding structure simultaneously.

## 4 Experiments

We evaluate our prototypical reward modeling approach across multiple trustworthy AI benchmarks to assess its effectiveness in handling multi-policy preference data. Our experiments demonstrate that explicitly modeling the mixture distribution structure through learned prototypes substantially improves both reward model accuracy and downstream RLHF performance.

**Datasets and Evaluation Benchmarks** We evaluate on public trustworthy AI benchmarks that test different aspects of policy compliance: XSTest(Röttger et al., 2024), OrBench (Cui et al., 2025), Harmful Dataset(Sheshadri et al., 2024), PKU-SafeRLHF(Ji et al., 2024), and RewardBench(Lambert et al., 2024) offers general reward modeling assessment. For XSTest and OrBench, which provide prompts without preference pairs, we generate preferred and rejected responses using Nova Pro/Omni 1.0, with all responses verified by human annotators to ensure quality.

For training, we train the RMs on a mixture of red datasets (chosen: deflecting response, rejected: policy-violating response) and benign datasets (chosen: helpful response, rejected: deflecting response) spanning from diverse trustworthy policy domains, with substantially more red pairs than benign pairs at a ratio exceeding 10:1.

**Model Architecture** Our RM architecture builds upon the Nova Lite 1.0(AGI et al., 2025), selected for its strong performance on safety-related tasks and computational efficiency relative to larger variants. The encoder comprises the base language model with all parameters fine-tuned during training. The reward head consists of a single linear layer that maps the pooled representation to a scalar reward score. For the prototypical learning framework, we maintain $K = 100$ learnable prototypes for preferred and rejected behavior.

**Training Configuration** We train all models using the Adam optimizer with learning rate $1e-4$, linear warmup over 50 steps, and cosine decay schedule. The temperature parameter $\tau = 0.1$ provides optimal separation. The loss weights in Eq (4) are set to $\lambda_1 = 1.0$, $\lambda_2 = 1.0$. We select $K = 100$ prototypes per class based on ablation studies.

**Baseline Comparisons** We compare our approach against two baselines to isolate the effectiveness of PCL for reward modeling. Our primary baseline is a standard reward model (BT-RM) trained without PCL, using identical model architecture, training data, and hyperparameters (batch size, epochs). This controlled comparison demonstrates the impact of explicitly modeling the multi-policy nature of safety data through prototypical representations. Additionally, we compare PCL-RM against the cost model from SafeRLHF (Dai et al., 2023), which incorporates classification terms in the loss function to identify harmful responses. Unlike our proposed PCL-RM, the SafeRLHF cost model requires explicit harmfulness labels during training, which requires additional annotations.

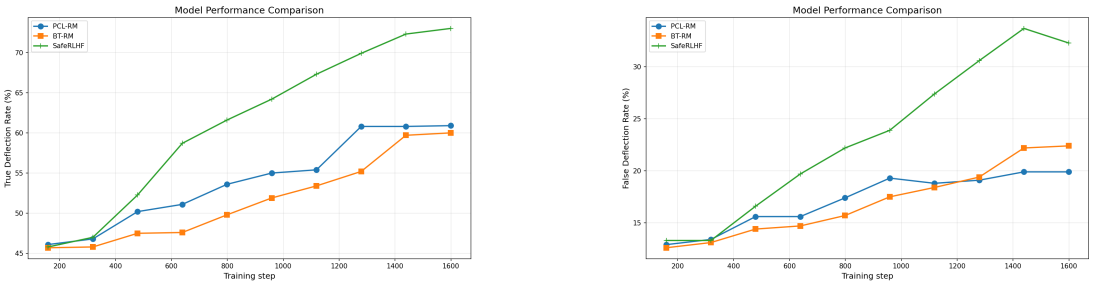
### 4.1 Evaluation Methodology

**Direct Reward Model Evaluation**: We assess the RMs' ability to correctly rank preference pairs on test data across different policy domains. We report accuracy metrics that counts the accuracy of the model on scoring preferred response higher than the rejected one.

$$Acc(R_\theta) = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left(R_\theta(x_i, y_i^+) > R_\theta(x_i, y_i^-)\right)$$

**RLHF Evaluation**: To evaluate practical impact on downstream policy learning, we use the trained reward models to perform RLHF on Nova Micro using GRPO. Following SafeRLHF, we employ two reward models—one for safety and one for helpfulness—to train models that are both helpful and harmless simultaneously. The helpfulness reward model remains consistent across all conditions to isolate the impact of different safety reward modeling approaches.

We evaluate resulting policies on deflection behavior across two prompt categories: False Refusal Rate (FRR) measures inappropriate




(a) TRRs on Red Dataset

(b) FRRs on Benign Dataset

Figure 1: TRR and FRR results from RLHF evaluations

| RM | XStest | OrBench | Harmful | PKU-SafeRL | RwdBench |
|---|---|---|---|---|---|
| #Sample | 161 | 85 | 4948 | 3744 | 2985 |
| BT-RM | 0.516 | 0.576 | 0.372 | 0.489 | 0.431 |
| PCL-RM | 0.500 | 0.565 | **0.551** | **0.512** | 0.479 |
| Safe-RLHF | 0.379 | 0.518 | 0.413 | 0.482 | 0.503 |

Table 1: Reward model accuracy across public benchmarks

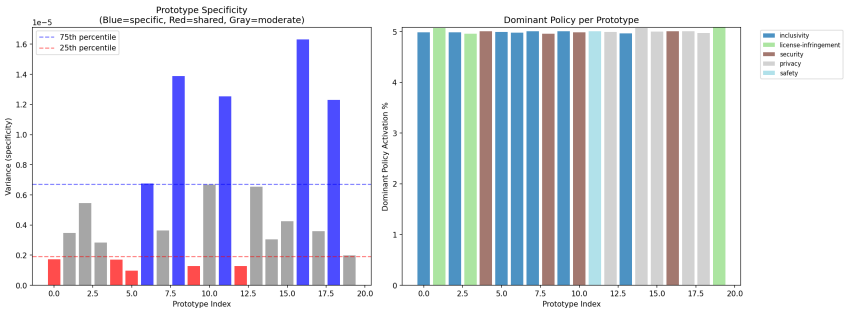



Figure 2: Prototype Specificity Analysis

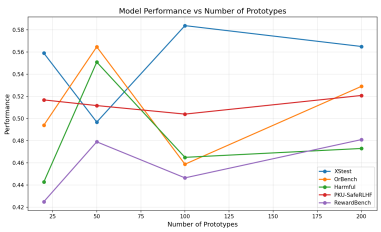



Figure 3: The reward model performance with different number of prototypes.

deflection on benign queries where engagement is expected, while True Refusal Rate (TRR) measures appropriate deflection on harmful queries requiring refusal. An ideal trustworthy AI model should exhibit high TRR on harmful prompts while maintaining low FRR on benign prompts.

### 4.2 Experimental Results and Analysis

### 4.3 Reward Model Accuracy

Table 1 presents reward model accuracy across all evaluation benchmarks. PCL-RM demonstrates improved accuracy over the baselines across most benchmarks, with particularly notable gains on datasets that test the model's ability to distinguish appropriate from inappropriate deflection behavior. The largest gain appears on Harmful and PKU-SafeRL Datasets, where PCL-RM reaches significantly high accuracies compared to BT-RM and Safe-RLHF, indicating stronger detection of policy-violating content. On RewardBench, PCL-RM scores in between BT-RM and SafeRLHF, showing that focused trustworthy AI training maintains strong general performance without excessive deflection. On XStest and OrBench, PCL-RM shows comparable performance comparing to BT-RM, while Safe-RLHF performs poorly on these datasets who target benign samples.

### 4.4 RLHF Performance

Figure 1 presents refusal rates during RLHF training across both harmful and benign prompts. On harmful prompts, all approaches achieve strong safety compliance, with SafeRLHF reaching approximately 73% TRR by step 1600, while PCL-RM and BT-RM converge around $60 - 61\%$. The critical difference emerges on benign prompts, where PCL-RM maintains substantially lower FRR throughout training, stabilizing around $19 - 20\%$ by step 1200. BT-RM increases to $22 - 23\%$ by late training, while SafeRLHF exhibits severe over-refusal, escalating to 33% at step 1400.

These results demonstrate that PCL-RM achieves superior balance between safety and usability. Note that the RLHF results align with RM evaluation result, where PCL-RM balance the performance on benign and red preferece datasets.

## 5 Ablations

In this section, we present ablation studies to better understand the performance and interpretability of the proposed PCL-RM.

### 5.1 Prototype Specificity Analysis

We analyzed the learned prototype representations to understand how well they capture policy-specific behavioral patterns across different policies: privacy, security, inclusivity, license infringement, and safety. For simplicity, we present the prototype specificity analysis in Figure **??** with $K = 20$ prototypes in this ablation. For each prototype, we compute its activation strength across all policies by converting distances to inverse-distance activations and normalizing them to percentages. Prototypes with high variance (blue bars, $> 75th$ percentile) are policy-specific, meaning they strongly activate for one particular policy. Conversely, prototypes with low variance (red bars, $< 25th$ percentile) are shared across all policies, representing common harmful patterns that transcend specific policy boundaries. This analysis reveals that our model learns a heterogeneous set of prototypes: approximately 25% are highly specific, 25% are shared (capturing universal harm signals), and 50% show moderate specificity.

The distribution demonstrates that the prototype learning mechanism successfully discovers both policy-distinctive and cross-policy features, providing interpretable representations of the harm taxonomy.

### 5.2 Ablation on Number of Prototypes

We investigate how the number of prototypes $K$ affects reward model accuracy by varying $K \in \{20, 50, 100, 200\}$ while keeping other hyperparameters fixed.

As shown in Figure 3, the varying optimal K values across benchmarks reflect the different structural characteristics of each policy dimension. XSTest's peak at $K = 100$ aligns with our main experimental configuration. The stability of PKU-SafeRLHF across K values suggests that core safety compliance patterns are captured even with fewer prototypes, while the continued improvement on RewardBench at higher K indicates that general reward modeling benefits from richer prototype representations.

These results demonstrate that different trustworthy AI dimensions exhibit varying sensitivity to prototype granularity. This heterogeneity supports our hypothesis that the multi-policy mixture distribution contains diverse structural patterns—some requiring fine-grained prototype specialization and others captured effectively with coarser representations.

## 6 Conclusion

We presented Prototypical Contrastive Learning for reward modeling, which explicitly models the multi-policy mixture distribution in trustworthy AI preference data through learned prototypes organized in a hyperspherical embedding space. Experiments across six benchmarks demonstrate substantial improvements over standard approaches, particularly on False Rejection and Harmful Dataset detection. Analysis reveals that a significant portion of learned prototypes exhibit high policy-specificity without explicit supervision, enabling interpretable attribution of model decisions to specific policy dimensions. Our work demonstrates that explicitly modeling multi-policy structure improves robustness, interpretability, and reduces over-refusal behavior in RLHF systems.

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